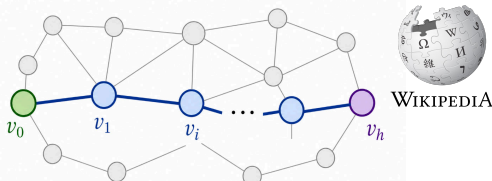


(a) High-Quality VQA Construction

1 Wikipedia Path Sampling

Constrained random walk on Wikipedia graph



$$P = (v_0 \rightarrow v_1 \rightarrow \dots \rightarrow v_h)$$

Anchor v_0 Bridge nodes $v_1, \dots, v_{h-1}$ Answer v_h

Constraints $\mathcal{C}$

- Skip disambiguation & list pages
- No cycles
- Hub avoidance (in-degree $> \tau_{hub}$)
- v_0 must be visually groundable



Accepted VQA Samples



Multi-hop VQA Pool ($\mathcal{I}, q, a$)

2 Canonical QA Generation

Extract short answer A from answer node v_h



Generate canonical question $q_{\dagger}$



3 Fuzzy Entity Rewriting

Progressively rewrite entities to obtain fuzzy question q_f



$q_{\dagger} \rightarrow \dots \rightarrow q_f$

Three Invariants

- ✓ Answer invariance
- ✓ Uniqueness
- ✓ Non-leakage

5 Quality Control



Run automated and model-based quality checks



Masking (feed the image)



Uniqueness (answer is unique)



Visual Relevance (image is useful)



Non-triviality (multi-hop reasoning)



4 Anchor Visual Grounding

Search representative Image I for anchor v_0 from Wikipedia



“the domed white building in the image”

Replace anchor mention with a visual referring expression in q_f

(b) Filtering and Enhancement

Staged Filtering

Filter A: Tool-Free Model

Discard if answer without tools



Filter B: One-call ImageSearch

Discard if a single image search already solves it



Goal: retain only tool-demanding, non-trivial samples



Filtered VQA Pool

Enhanced Subset (10%)



Select 10% of filtered samples

Degrade Images



Blur

Downsampling

Perspective Distortion

Enhance with Tools



Sharpen SuperResolution PerspectiveCorrect Exercises think-with-image behavior.

(c) Expert Trajectory Synthesis

Multi-turn Trajectory Synthesis



Expert Model: Claude Opus 4.6

Interact with real environment E via tool set $T = T_v \cup T_r$



Browse



Page-Visit



Search



Python



Think



Tool



Observation



Think



Tool



Answer



Raw Expert Rollouts

Rejection Sampling

Answer Correctness Judge



Final answer must match ground truth A



Process-level Judge

At least one tool call

✓ Logical consistency with tool observations

✓ No ineffective repetition



High-Quality Expert Trajectories

36592 Trajectories;
Avg 6.3 tool turns